



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 02

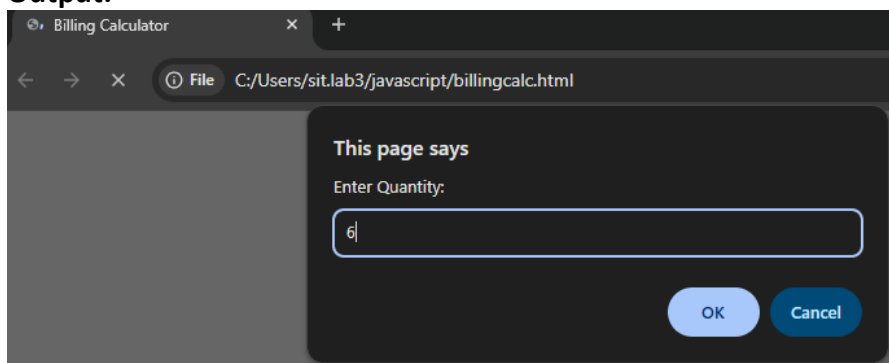
1. Title: Use var, let, const, and demonstrate basic features like template literals and destructuring, build a billing calculator with user input.

2. Software/Tools Required: HTML, VS Code.

3. Theory: JavaScript is a scripting language used to make web pages interactive and dynamic. It allows developers to perform calculations, validate user input, display messages, and manipulate webpage content. In this experiment, a **Billing Calculator** is developed using JavaScript. The program accepts user inputs such as customer name, product name, quantity, and price per item through dialog boxes (prompt()). It then calculates the total bill and displays a formatted billing receipt on the webpage. The program demonstrates the following JavaScript features:

- **var** – Used to declare function-scoped variables. It can be redeclared and updated.
- **let** – Used to declare block-scoped variables. It can be updated but not redeclared within the same scope.
- **const** – Used to declare constants whose values cannot be reassigned.
- **Template Literals** – Enclosed within backticks (`), they allow embedding variables directly into strings using `\${}` for easier string formatting.
- **Destructuring** – A feature that extracts values from arrays or objects into separate variables, making the code cleaner and more readable.

4. Output:





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← ↻ 127.0.0.1:5500/index.html/mahimadugar

Billing Calculator

Billing Receipt

Customer: mahima

Product: laptop

Quantity: 3

Price per Item: Rs40000

Total: Rs120000

GST (18%): RS.21600.00

Final Bill: Rs.141600.00



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5. Case Study Title: Develop a webpage for an online shopping website using HTML, CSS, and HTML5 tags, and calculate the bill.

6. Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Online Shopping Bill Calculator</title>

<style>
body{
    font-family: Arial, sans-serif;
    background:#f2f2f2;
}

.container{
    width:500px;
    margin:30px auto;
    background:white;
    padding:20px;
    border-radius:10px;
    box-shadow:0 0 10px gray;
}

h2{
    text-align:center;
    color:darkblue;
}

label{
    display:block;
    margin-top:10px;
    font-weight:bold;
}

input,select{
    width:100%;
    padding:8px;
    margin-top:5px;
}
```



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```
button{
  width:100%;
  padding:10px;
  margin-top:20px;
  background:green;
  color:white;
  border:none;
  font-size:16px;
  cursor:pointer;
}
```

```
button:hover{
  background:darkgreen;
}
```

```
#result{
  margin-top:20px;
  padding:15px;
  background:#e8f5e9;
  border-radius:5px;
}
```

```
</style>
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h2>Online Shopping Bill Calculator</h2>
```

```
<label>Customer Name</label>
<input type="text" id="name">
```

```
<label>Mobile Number</label>
<input type="text" id="mobile">
```

```
<label>Invoice Number</label>
<input type="text" id="invoice">
```

```
<label>Product Name</label>
<input type="text" id="product">
```

```
<label>Quantity (Kg)</label>
<input type="number" id="qty">
```

```
<label>Rate per Kg (₹)</label>
<input type="number" id="rate">
```



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```
<label>Discount (%)</label>
<input type="number" id="discount">

<label>GST (%)</label>
<input type="number" id="gst">

<label>Packing Charges (₹)</label>
<input type="number" id="packing">

<label>Payment Mode</label>
<select id="payment">
  <option>Cash</option>
  <option>Card</option>
  <option>UPI</option>
</select>

<label>Membership</label>
<select id="member">
  <option>Yes</option>
  <option>No</option>
</select>

<button onclick="calculateBill()">Generate Bill</button>

<div id="result"></div>

</div>

<script>

function calculateBill(){

  // String Data Types
  var customerName = document.getElementById("name").value;
  var mobile = document.getElementById("mobile").value;
  var invoice = document.getElementById("invoice").value;
  var product = document.getElementById("product").value;

  // Number Data Types
  let qty = Number(document.getElementById("qty").value);
  let rate = Number(document.getElementById("rate").value);
  let discount = Number(document.getElementById("discount").value);
  let gst = Number(document.getElementById("gst").value);
  let packing = Number(document.getElementById("packing").value);

  // Constant
  const payment = document.getElementById("payment").value;
```



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```
// Boolean Data Type
let membership = document.getElementById("member").value=="Yes";

// Undefined and Null
let remarks;
let coupon = null;

let amount = qty * rate;

let discountAmount = amount * discount / 100;

let afterDiscount = amount - discountAmount;

let gstAmount = afterDiscount * gst / 100;

let total = afterDiscount + gstAmount + packing;

if(membership){
    total = total - (total * 5 / 100);
}

document.getElementById("result").innerHTML = `
<h3>Invoice</h3>

Customer Name : ${customerName}<br>
Mobile Number : ${mobile}<br>
Invoice Number : ${invoice}<br>
Product Name : ${product}<br><br>

Quantity : ${qty} Kg<br>
Rate : ₹${rate}<br>
Amount : ₹${amount.toFixed(2)}<br>
Discount : ₹${discountAmount.toFixed(2)}<br>
GST : ₹${gstAmount.toFixed(2)}<br>
Packing Charges : ₹${packing}<br>
Membership : ${membership}<br>
Payment Mode : ${payment}<br><br>

<h3>Total Bill : ₹${total.toFixed(2)}</h3>

<br>

<b>Primitive Data Types Used</b><br>
String : ${typeof customerName}<br>
Number : ${typeof qty}<br>
Boolean : ${typeof membership}<br>
Undefined : ${typeof remarks}<br>
Null : ${coupon}
```



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```
;
}

</script>

</body>

</html>
```

7. Output:

← → ↻ C:/Users/Hp/Javascript/billing.html/mahimadugar

Online Shopping Bill Calculator

Customer Name	Mahima Dugar
Mobile Number	8208239478
Invoice Number	123
Product Name	tea leaves
Quantity (Kg)	3
Rate per Kg (₹)	150
Discount (%)	5
GST (%)	2.5
Packing Charges (₹)	12
Payment Mode	Cash
Membership	Yes
Generate Bill	



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Generate Bill

Invoice

Customer Name : Mahima Dugar
Mobile Number : 8208239478
Invoice Number : 123
Product Name : tea leaves

Quantity : 3 Kg
Rate : ₹150
Amount : ₹450.00
Discount : ₹22.50
GST : ₹10.69
Packing Charges : ₹12
Membership : true
Payment Mode : Cash

Total Bill : ₹427.68

Primitive Data Types Used

String : string
Number : number
Boolean : boolean
Undefined : undefined
Null : null

- 5. Result/Conclusion:** The Billing Calculator was successfully implemented using JavaScript. The program accepts user input, performs the billing calculation, and displays the receipt correctly. It successfully demonstrates the use of **var**, **let**, **const**, **template literals**, and **destructuring** while providing a simple and interactive billing application.